Week 9: Select the Winning Model

1. Validation Results for All Nine Models

Six of the nine models score within 0.013 of each other on validation PR-AUC (0.722 to 0.735). The three most flexible single models, MLP Variation 3, gradient boosting and the random forest, score clearly lower and overfit the most.

All nine models were scored on the same 15 train/validation splits used in Weeks 6 to 8: 5-fold stratified cross-validation repeated 3 times on the 1,608 training customers. Each value is the average over the 15 validation folds. PR-AUC is the main metric because only 14.6% of customers respond. Log loss is the error metric: it measures how far the predicted probabilities are from what actually happened, and lower is better. F1 uses a 0.5 probability cutoff, as in earlier weeks.

Table 1. Training and validation metrics for all nine models

Model	Training PR-AUC	Validation PR-AUC	Gap (PR-AUC)	Validation ROC-AUC	Validation F1	Validation log loss
W6 Logistic regression	0.772	0.732	0.040	0.930	0.633	0.227
W7 MLP Variation 1	0.779	0.725	0.054	0.928	0.596	0.229
W7 MLP Variation 2	0.867	0.722	0.145	0.927	0.632	0.227
W7 MLP Variation 3	1.000	0.678	0.322	0.914	0.623	0.567
W6 Gradient boosting	0.885	0.693	0.193	0.910	0.560	0.248
W6 Random forest	0.968	0.650	0.317	0.900	0.390	0.272
W8 Stack Variation 1	0.779	0.731	0.048	0.929	0.629	0.240
W8 Stack Variation 2	0.813	0.735	0.079	0.931	0.625	0.239
W8 Stack Variation 3	0.799	0.726	0.073	0.929	0.619	0.229

The gap column is training minus validation PR-AUC; a large gap means the model memorized its training customers. Logistic regression has the smallest gap of all nine (0.040) and the lowest validation log loss (0.227, tied with MLP Variation 2). Stack Variation 2 has the highest validation PR-AUC (0.735), only 0.003 ahead of logistic regression.

2. The Winning Model: Tuned Logistic Regression

We selected the tuned logistic regression from Week 6 (C = 3, L2 penalty, no class weights). It is statistically tied with the best model, it is the simplest of the nine, and it overfits the least.

Selection rule. Validation scores vary from fold to fold, so a small lead in the mean can be noise. We used the one standard error rule: find the model with the highest mean validation PR-AUC, then choose the

least complex model whose mean is within one standard error of it. The standard error is the fold standard deviation divided by the square root of 15.

- The highest mean validation PR-AUC is Stack Variation 2 at 0.735, with a standard error of 0.015, so the threshold is 0.719.
- Six models clear the threshold: logistic regression, MLP Variations 1 and 2, and Stack Variations 1 to 3.
- Logistic regression is the least complex of the six, so it is selected.

The lead of the more complex models is noise. Stack Variation 2 beats logistic regression on only 8 of the 15 folds, by an average of 0.002 PR-AUC, which is about one seventh of a standard error. Stack Variation 1 beats it on 7 of 15 folds and averages 0.001 lower. The more flexible single models lose clearly: the random forest beats logistic regression on 2 of 15 folds and gradient boosting on 3 (Appendix A).

Why logistic regression performed best.

1. **The dataset is small for flexible models.** The training set has 1,608 customers but only 234 responders, about 9 per feature. Models with many parameters have too few positive examples to learn real patterns, so they memorize individual customers instead. MLP Variation 3 learns 3,905 weights and reaches a perfect training score (PR-AUC 1.000) but only 0.678 on validation.
2. **The signal is mostly additive.** The Week 5 features, such as the count of past campaigns accepted, spending and recency, already capture the main patterns in a form a linear model can use. The small MLP ranks customers almost identically to logistic regression (Spearman correlation 0.975, Week 8), so its extra flexibility finds nothing new.
3. **Combining models adds little.** Stacking only helps when its base models make different mistakes. Gradient boosting is the only base model that ranks customers differently, but it is also the weakest (0.693), so what it adds is partly noise.
4. **Its probabilities are the most reliable.** Logistic regression has the lowest validation log loss (0.227). This matters for the business case, because contact decisions depend on the predicted probability of response, not only on the ranking.

Logistic regression is also the cheapest to train and run, and the easiest to explain to a marketing team, since each coefficient shows whether a feature raises or lowers the chance of response.

3. Bias-Variance Tradeoff

The chart shows that this prediction task is limited by variance, not bias: added flexibility lowers training error but does not lower validation error, and for the most flexible models it raises it.

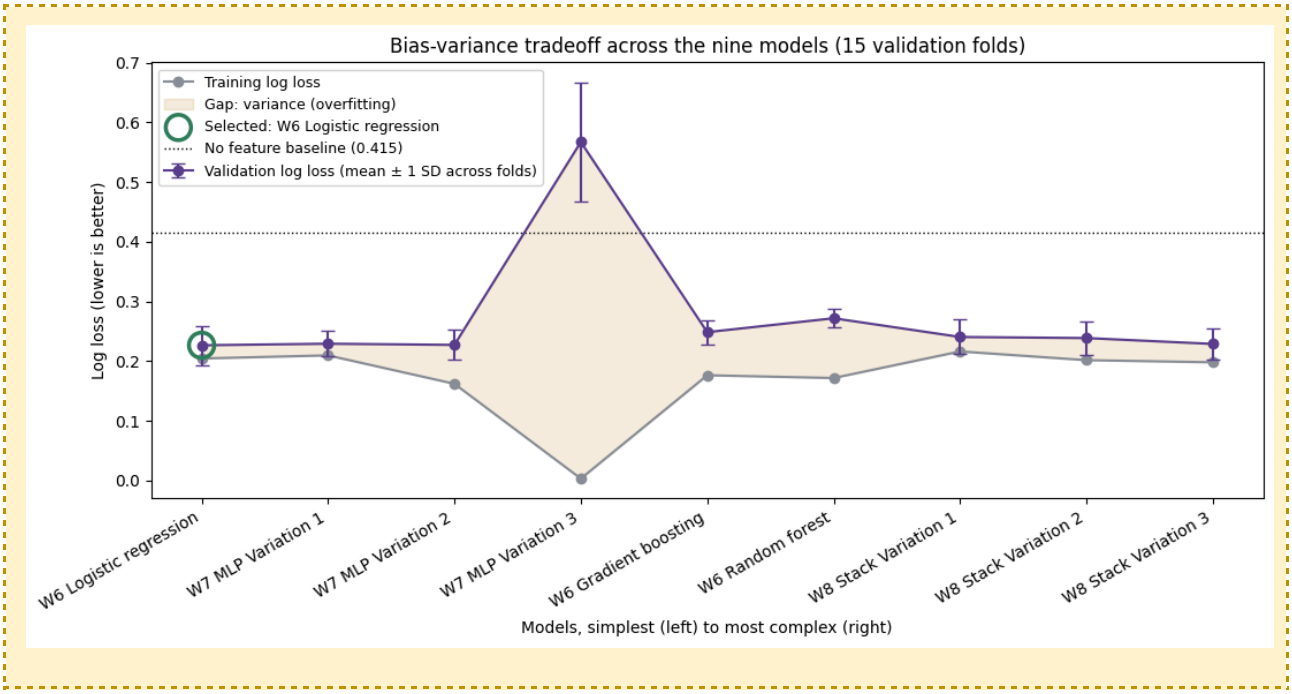


Figure 1. Training and validation log loss (lower is better) for the nine models, ordered from simplest (left) to most complex (right). Error bars show one standard deviation across the 15 validation folds; the shaded area is the gap between training and validation error. The dotted line is a model with no features that predicts the 14.6% response rate for every customer.

How the models are ordered. The linear model comes first, then the neural networks from smallest to largest (233, 465 and 3,905 weights), then the tree ensembles (gradient boosting with 200 shallow trees, then the random forest with 200 unrestricted trees), and last the stacking ensembles, because each one contains several of the other models.

What the chart shows.

- **Training error falls as models get more flexible, so bias drops.** Training log loss goes from 0.204 for logistic regression to 0.003 for MLP Variation 3, which predicts its training customers almost perfectly.
- **Validation error does not follow.** The three simplest models (logistic regression and MLP Variations 1 and 2) sit at about the same validation log loss (0.227 to 0.229). Validation error then rises for the flexible single models: 0.248 for gradient boosting, 0.272 for the random forest and 0.567 for MLP Variation 3.
- **The gap grows with flexibility, so variance rises.** The gap between validation and training log loss is 0.022 for logistic regression and 0.020 for MLP Variation 1, against 0.072 for gradient boosting, 0.100 for the random forest and 0.564 for MLP Variation 3.

- **The most flexible network is worse than no model at all on this metric.** MLP Variation 3's validation log loss (0.567) is higher than the no-feature baseline (about 0.41). It still ranks customers reasonably (PR-AUC 0.678), but its probabilities are overconfident, because it memorized its training customers.
- **The stacking ensembles keep variance low but do not reduce error.** Their gaps are small (0.024 to 0.037), but their validation log loss (0.229 to 0.240) is no better than logistic regression's.

What this means for the task. The best validation error is reached by the simplest model, and every step up in flexibility either matches it or does worse. With 234 responders, the extra capacity of the flexible models is spent memorizing noise. Better performance is more likely to come from more customer data or more informative features than from more complex models. The Week 7 learning curve supports this: validation PR-AUC was still rising slowly when the full training set was used.

4. Test Set Performance

On the 412 test customers, the selected model reached a PR-AUC of 0.665, lower than its validation score of 0.732 but inside the range we predicted before scoring the test set. It ranks customers well enough to target a campaign, but whether targeting pays depends on the cost of each contact.

Prediction made before scoring the test set. The mean validation PR-AUC (0.732) was the estimate of test PR-AUC. The expected range, 0.615 to 0.849, scales the fold-to-fold spread to the size of the test set. The test set was then scored once, with the model retrained on all 1,608 training customers.

Table 2. Test set metrics for the selected model

Metric	Test value
PR-AUC	0.665 (95% bootstrap interval 0.559 to 0.766)
ROC-AUC	0.883
Precision (0.5 cutoff)	0.692
Recall (0.5 cutoff)	0.429
F1 (0.5 cutoff)	0.529
Log loss	0.281

What the test performance means for the task.

- **The model ranks customers far better than chance.** Random contact would give a PR-AUC equal to the response rate, 0.153. The model's 0.665 is more than four times that, and its highest-ranked customers almost all respond: precision stays at 100% until about a quarter of all responders are reached (Appendix B).

- **Targeting the top 40% captures most responders.** Contacting the 40% of test customers with the highest predicted probability reaches 87.3% of all responders, with a response rate of 33.5% against 15.3% overall, a 2.2 times lift.
- **The 0.5 cutoff is not the right decision rule.** At that cutoff the model finds only 43% of responders. In practice the marketing team would contact a fixed share of customers ranked by probability, such as the top 40%, rather than only those above 0.5.

Is the model good enough? For ranking customers, yes: it captures nearly nine in ten responders while contacting fewer than half of customers. Whether that saves money depends on the cost of a contact. Applying the test results to the business case from our project plan (2,000 customers, 15% response, $2 per contact, $150 per accepted offer):

Table 3. Business case: contact everyone vs. contact the top 40%

Strategy	Customers contacted	Responders reached	Revenue	Contact cost	Net value
Contact everyone	2,000	300	$45,000	$4,000	$41,000
Contact the top 40%	800	262	$39,285	$1,600	$37,685

At $2 per contact, contacting everyone earns about $3,300 more, because the 38 responders the model misses are worth $5,715 while the 1,200 contacts it saves cost only $2,400. Targeting pays off once a contact costs more than about $4.76 (the $5,715 divided by 1,200 contacts). It is therefore the better strategy for costlier channels such as direct mail or phone calls, when the contact budget or offer supply is limited, or when contacting uninterested customers causes unsubscribes. We are satisfied with the model as a ranking tool, and the business case should state this break-even condition.

5. Training, Validation and Test Performance of the Winning Model

Every metric declines from training to validation to test: a small drop from training to validation, then a larger one from validation to test.

Table 4. Selected model (tuned logistic regression) across the three datasets

Metric	Training (mean of 15 folds)	Validation (mean of 15 folds)	Test
PR-AUC	0.772	0.732	0.665
ROC-AUC	0.941	0.930	0.883
Precision	0.784	0.740	0.692
Recall	0.576	0.558	0.429
F1	0.664	0.633	0.529

Metric	Training (mean of 15 folds)	Validation (mean of 15 folds)	Test
Log loss (lower is better)	0.204	0.227	0.281

Training to validation: a small drop, so the model generalizes. PR-AUC falls by 0.040 and log loss rises by 0.023, among the smallest changes of the nine models. The model is not memorizing its training customers.

Validation to test: a larger drop, within what chance allows. PR-AUC falls by 0.067, ROC-AUC by 0.047 and recall by 0.129. Two explanations are possible, and the data cannot fully separate them:

1. **Chance.** The test set has 412 customers and only 63 responders, so one test score is noisy. The bootstrap interval for test PR-AUC (0.559 to 0.766) contains the validation estimate of 0.732, and the result falls inside the range predicted beforehand.
2. **A real difference between training and test customers.** Because every metric moved the same way, the test customers may differ somewhat from the training customers. Checking how the test set was created in Weeks 3 to 5 would show whether this is likely.

Precision held up better than recall. Precision fell from 0.740 to 0.692, while recall fell from 0.558 to 0.429. The customers the model is most confident about still respond at a high rate, but fewer test responders reach a predicted probability of 0.5. This is another reason to target a fixed top share of customers rather than use the 0.5 cutoff.

We did not switch models after seeing the test result. Choosing a different model based on test scores would turn the test set into a second validation set and make the reported performance overly optimistic. For planning, production performance should be expected closer to the test result, a PR-AUC of about 0.67, than to the validation estimate.

6. Conclusion

The tuned logistic regression from Week 6 is the final model. It is statistically tied with the best of the nine models on validation PR-AUC (0.732 against 0.735), has the lowest validation log loss (0.227), overfits the least, and is the simplest to run and explain. The bias-variance chart shows why the more complex models did not help: with only 234 responders, their extra flexibility memorized training customers instead of finding new patterns.

On the test set the model reached a PR-AUC of 0.665, inside the range predicted beforehand. Contacting its top 40% of customers reaches 87.3% of responders with a 2.2 times lift. The model is good enough for ranking customers; targeting saves money once a contact costs more than about $4.76, which the business case should reflect.

Appendix A: Fold-by-Fold Comparison with the Selected Model

Each model's validation PR-AUC compared with logistic regression's on the same 15 folds. A positive mean difference means the model scored higher on average.

Model	Mean difference vs logistic regression	Folds better than logistic regression (of 15)
W7 MLP Variation 1	-0.007	6
W7 MLP Variation 2	-0.010	7
W7 MLP Variation 3	-0.054	[fill in from the 9.3 output]
W6 Gradient boosting	-0.040	3
W6 Random forest	-0.082	2
W8 Stack Variation 1	-0.001	7
W8 Stack Variation 2	+0.002	8
W8 Stack Variation 3	-0.006	5

Appendix B: Test Set Precision-Recall Curve

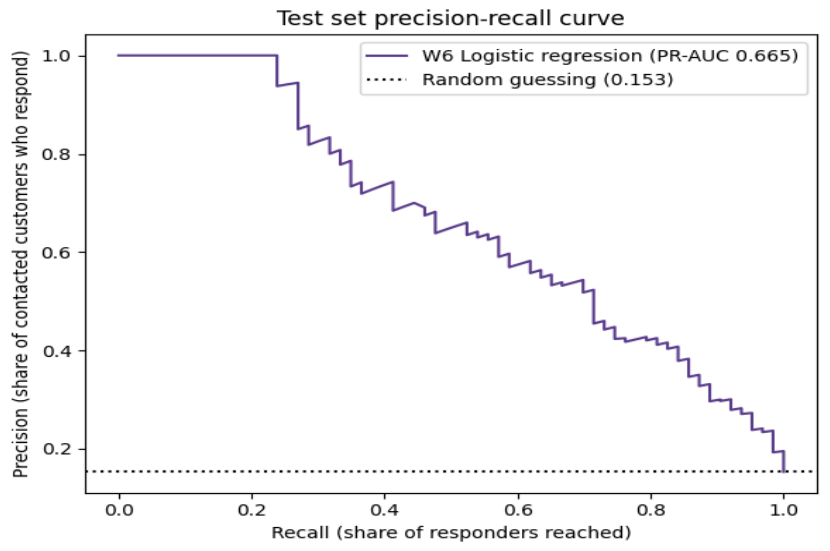


Figure 2. Precision and recall on the 412 test customers as the contact threshold changes. Precision stays at 100% until about a quarter of responders are reached, and stays well above the 15.3% random rate across most of the curve. The steps come from the small test set: each one is a single responder.

Appendix C: Settings of the Nine Models

Model	Week	Settings
Logistic regression	6	C = 3, L2 penalty, no class weights

Model	Week	Settings
MLP Variation 1	7	1 hidden layer of 8 units, alpha = 3.0 (233 weights)
MLP Variation 2	7	1 hidden layer of 16 units, alpha = 1.0 (465 weights)
MLP Variation 3	7	2 hidden layers of 64 and 32 units, alpha = 0.001 (3,905 weights)
Gradient boosting	6	200 trees, learning rate 0.05, max depth 2, subsample 0.8
Random forest	6	200 trees, no depth limit, at least 5 customers per leaf, no class weights
Stack Variation 1	8	Logistic regression + MLP Variation 1; meta-model C = 1.0
Stack Variation 2	8	Logistic regression + gradient boosting + MLP Variation 1; meta-model C = 1.0
Stack Variation 3	8	As Variation 2, plus the 27 original features (passthrough); meta-model C = 0.1

All MLPs use ReLU activation, the Adam solver, a learning rate of 0.001, up to 500 iterations and feature scaling inside the pipeline. All stacking meta-models are scaled logistic regressions trained on inner 5-fold predictions.